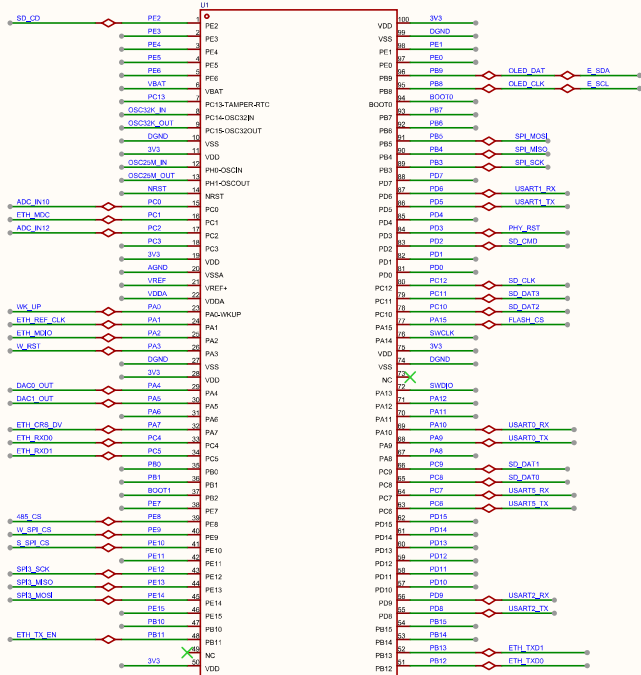
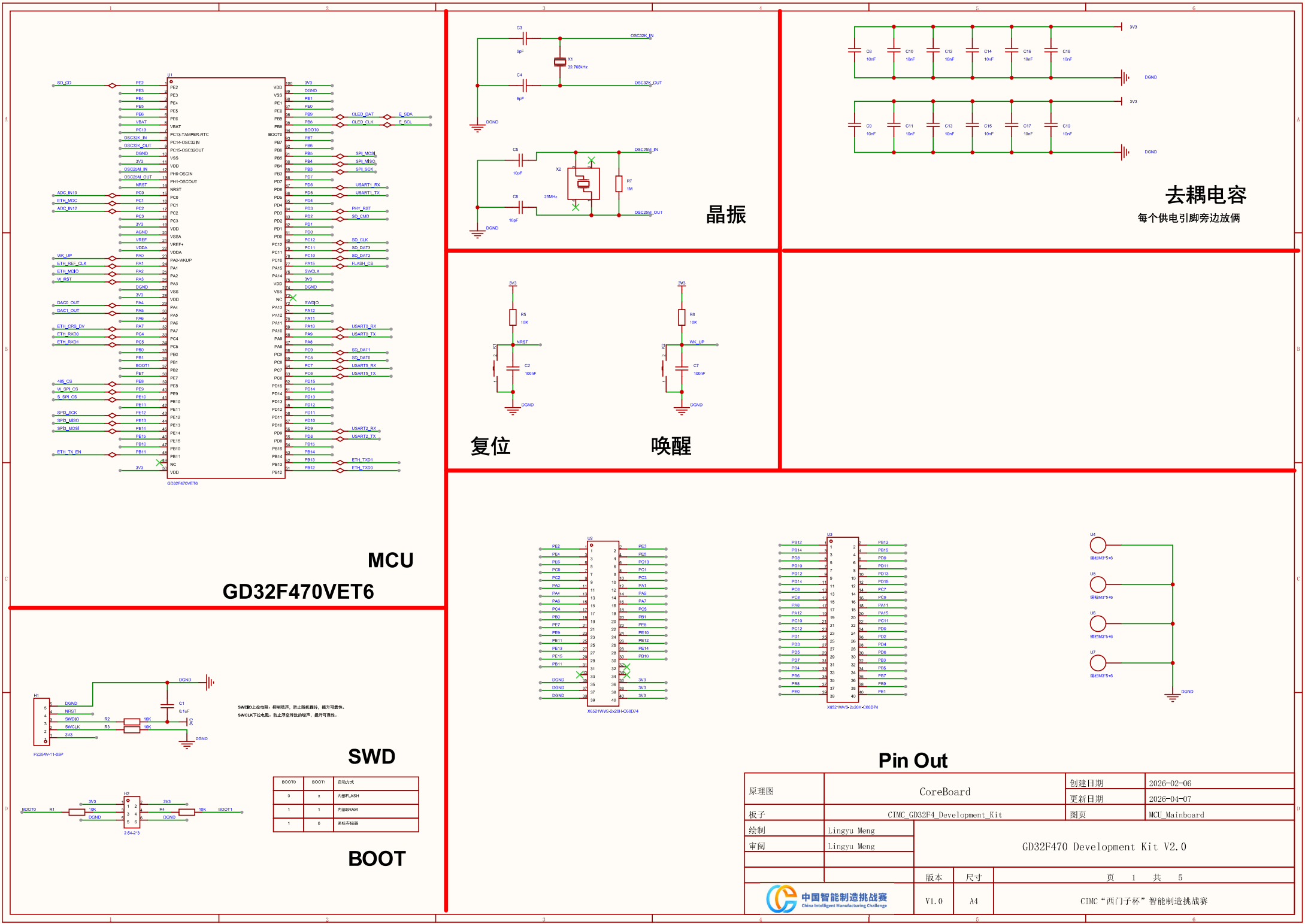
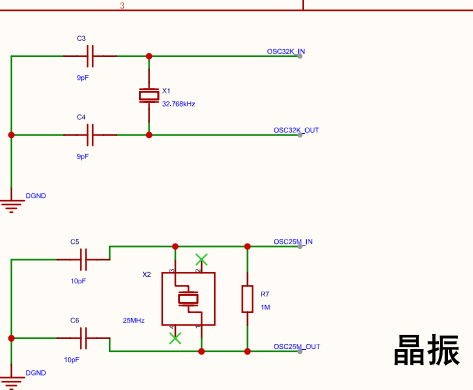
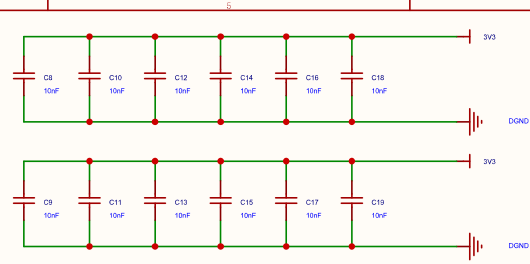




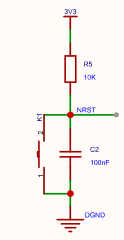
GD32F470VET6



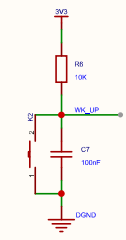
晶振



去耦电容
每个供电引脚旁边放俩

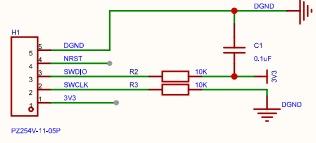


复位



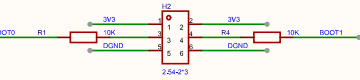
唤醒

MCU
GD32F470VET6



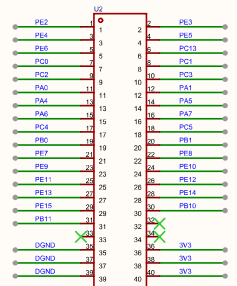
SWDIO上拉电阻：抑制噪声，防止随机翻转，提升可靠性。
SWCLK下拉电阻：防止浮空导致的噪声，提升可靠性。

SWD

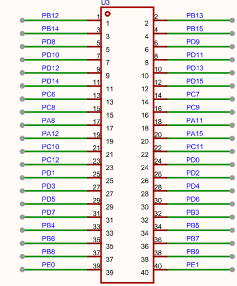


BOOT0	BOOT1	启动方式
0	x	内部FLASH
1	1	内部SRAM
1	0	系统存储器

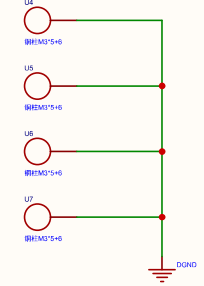
BOOT



X8521WVS-200H-C80D74



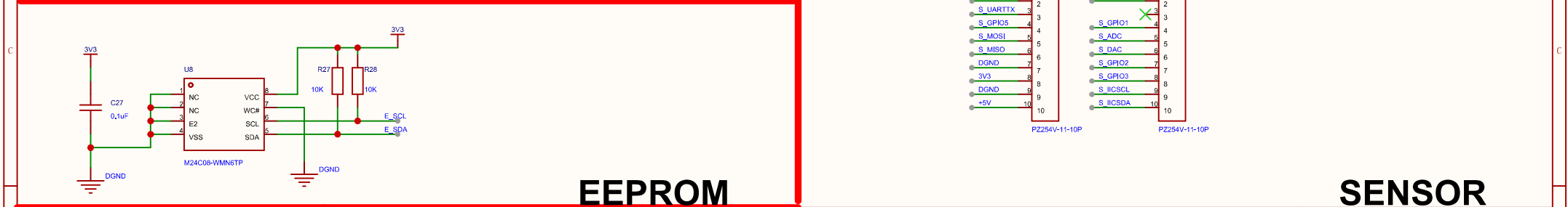
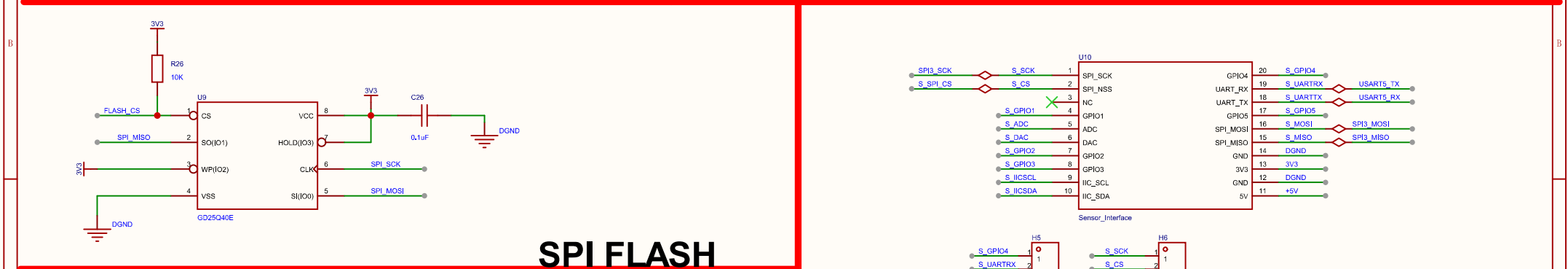
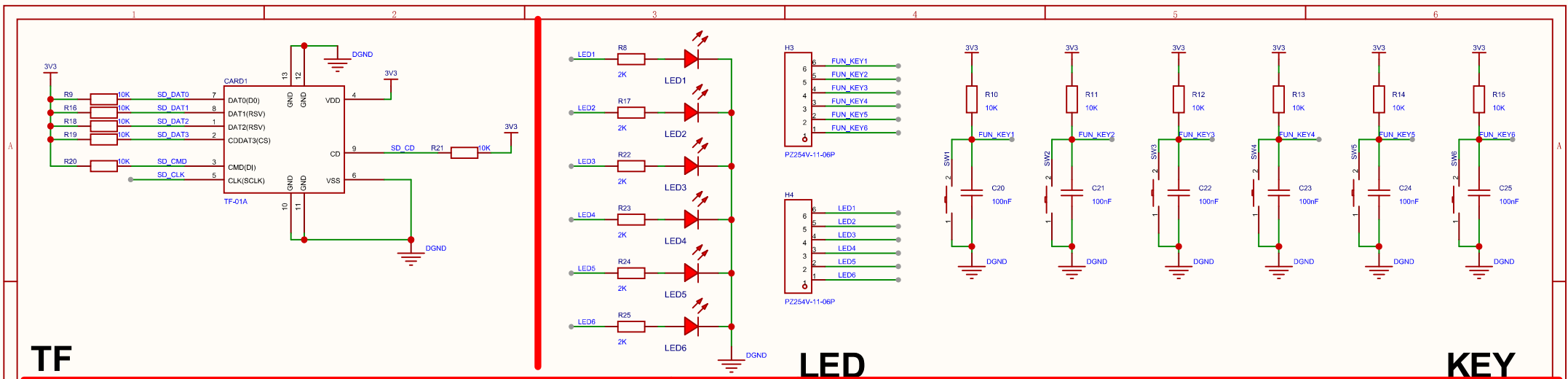
X8521WVS-200H-C80D74



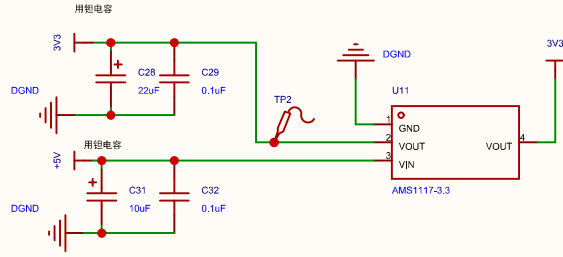
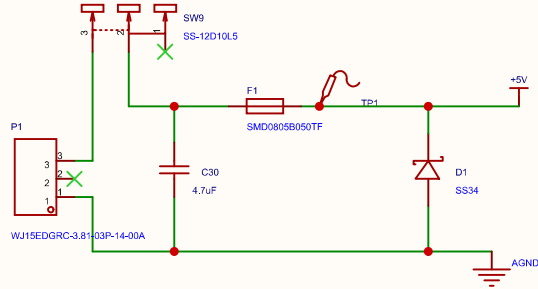
U4

Pin Out

原理图	CoreBoard	创建日期	2026-02-06			
		更新日期	2026-04-07			
板子	CIMC_GD32F4_Development_Kit		图页	MCU_Mainboard		
绘制	Lingyu Meng	GD32F470 Development Kit V2.0				
审阅	Lingyu Meng					
		版本	尺寸	页 1 共 5		
	中国智能制造挑战赛 China Intelligent Manufacturing Challenge	V1.0	A4	CIMC “西门子杯” 智能制造挑战赛		

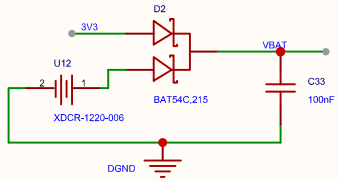


5V输入

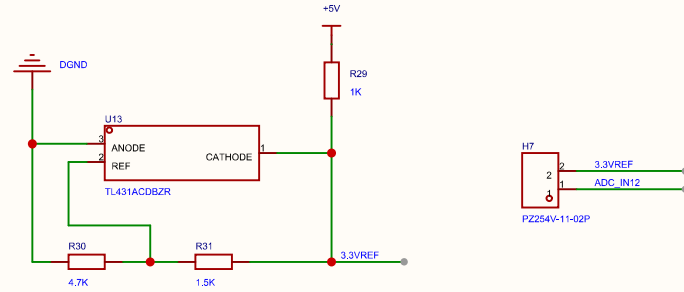


5V - 3V3

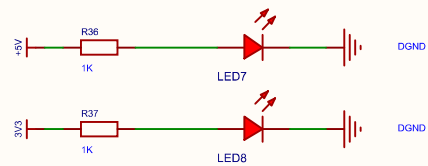
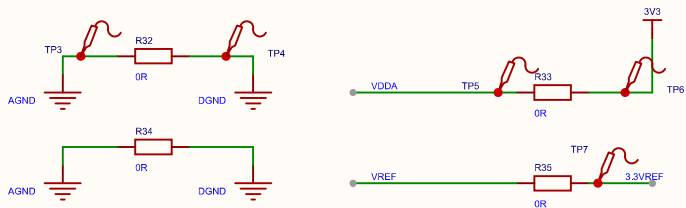
Vbat



$$V_{REF} = 2.5 * (1 + R1(1.5k)/R2(4.7k)) = 3.2979V$$



ADC基准



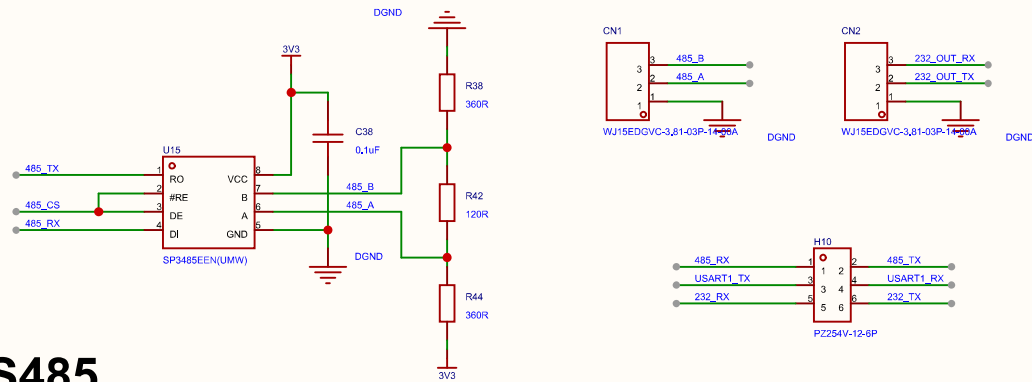
5V 电源灯



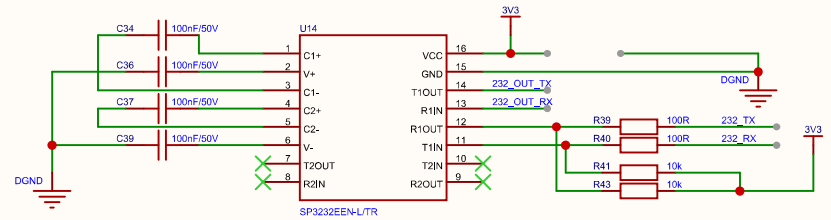
3.3V 电源灯

原理图	CoreBoard		创建日期	2026-02-06
板子	CIMC_GD32F4_Development_Kit		更新日期	2026-04-07
绘制	Lingyu Meng	GD32F470 Development Kit V2.0		
审阅	Lingyu Meng			
		版本	尺寸	页 3 共 5
		V1.0	A4	CIMC “西门子杯” 智能制造挑战赛

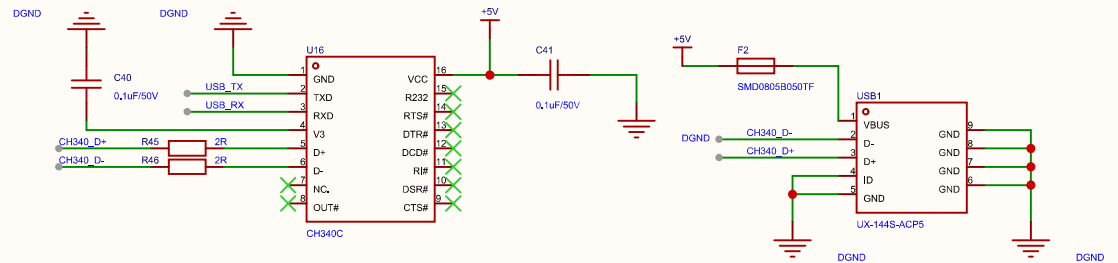
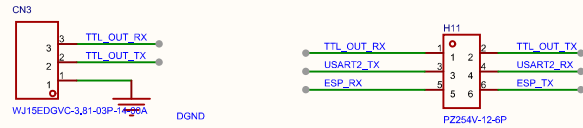
RS485



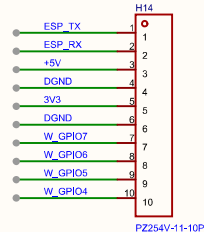
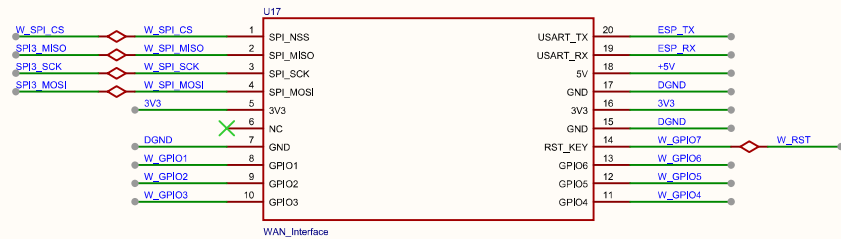
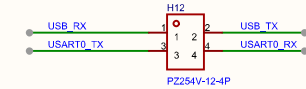
RS232 USART1



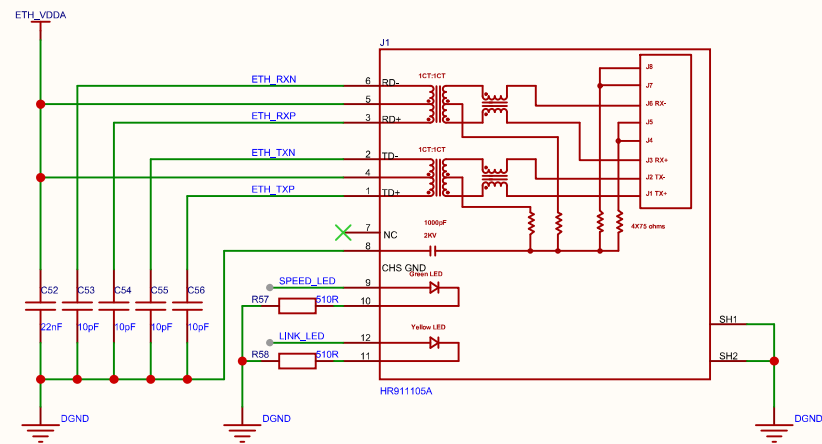
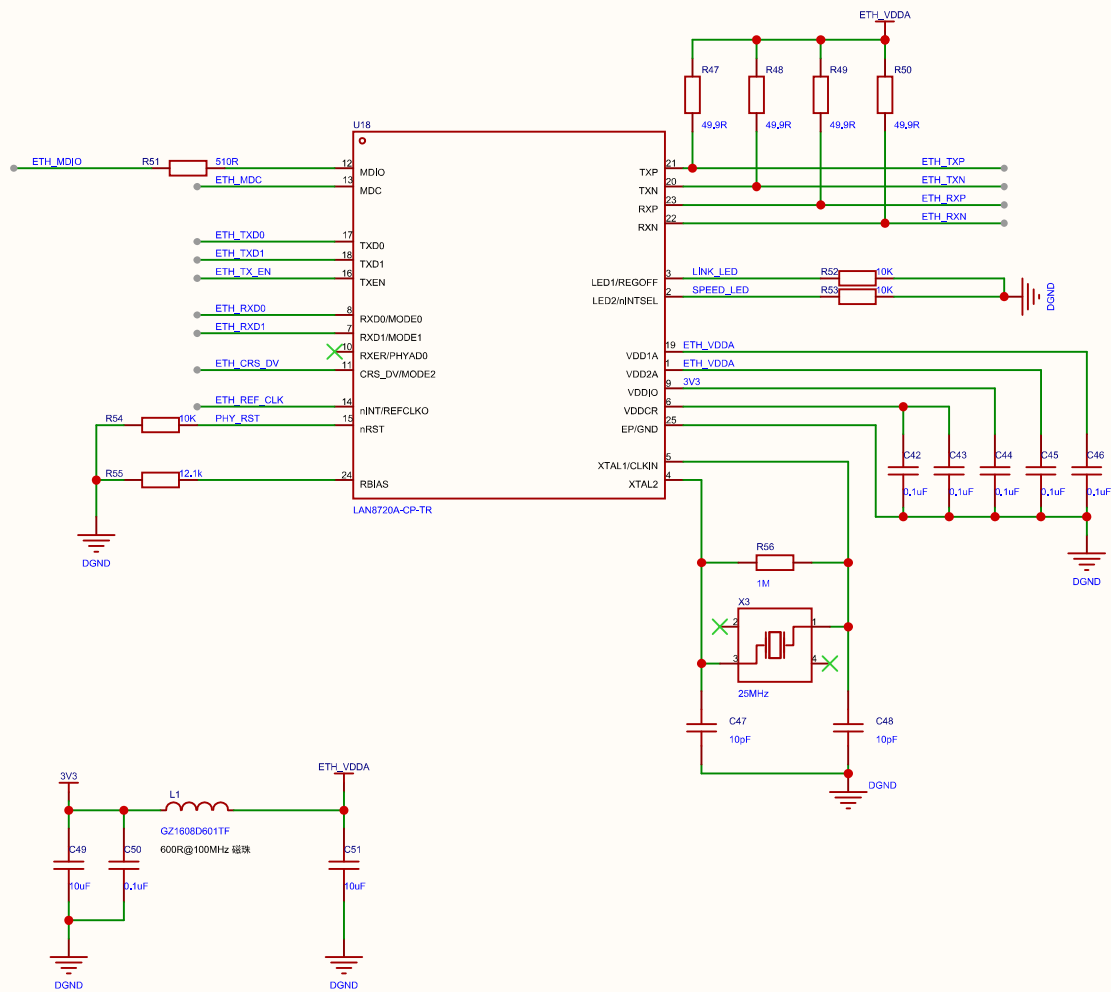
USART2



USB USART0



原理图	CoreBoard	创建日期	2026-02-06
		更新日期	2026-04-07
板子	CIMC_GD32F4_Development_Kit	图页	commInterface
绘制	Lingyu Meng	GD32F470 Development Kit V2.0	
审阅	Lingyu Meng		
		版本	尺寸
		V1.0	A4
		页 4 共 5	
		CIMC “西门子杯” 智能制造挑战赛	



原理图	CoreBoard		创建日期	2026-02-06
			更新日期	2026-02-06
板子	CIMC_GD32F4_Development_Kit		图页	Ethernet
绘制	GD32F470 Development Kit V2.0			
审阅				
		版本	尺寸	页 5 共 5
		V1.0	A4	嘉立创EDA